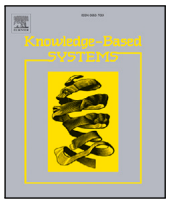
Knowledge-Based Systems 347 (2026) 116267



Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys





Automated data synthesis and retrieval-augmented generation for legal large language models

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ARTICLE INFO

Keywords:
Legal LLMs
Retrieval-augmented generation
Data synthesis
LLM-as-a-judge
ELO-ranking

ABSTRACT

The application of large language models (LLMs) in the legal domain faces two critical challenges: data scarcity and the need for trustworthy outputs. Legal data is limited and often lacks the quality and structure necessary to train robust AI systems. Furthermore, legal tasks demand high accuracy and reliability, making it essential for AI systems to generate trustworthy content. To address these challenges, we present the first automated framework for generating high-quality retrieval-augmented generation (RAG) data tailored specifically for legal LLMs. Our approach involves synthesizing high-quality question–answer pairs relevant to the legal domain. Each generated pair is then evaluated by an LLM-as-a-Judge, which classifies them as either good or bad based on their quality. Subsequently, we apply an ELO-Ranking technique to rank all pairs, ensuring that the most reliable and accurate pairs are selected for further use. These high-quality pairs are then employed to train both the retrieval and generation modules of a RAG model, addressing the issue of data scarcity and improving the overall performance of legal LLMs. Experimental validation on the LawBench benchmark demonstrates that our approach significantly enhances model performance, providing a reliable solution for training trustworthy legal AI models with RAG-based techniques.

1. Introduction

With the rapid development of Artificial Intelligence (AI) and Natural Language Processing (NLP) technologies, large language models have demonstrated significant capabilities across various domains, including text generation, intelligent question answering, and document analysis. In the legal domain, the application of LLMs has shown great potential, particularly in automating the generation of legal documents, legal consultation, and case analysis. However, despite the promising results, the application of these technologies within the Chinese legal field faces several challenges, with one of the most critical being the scarcity of high-quality training data.

Legal data in China are relatively limited, especially standardized data and suitable for training AI systems. Legal resources in China consist mainly of statutes, judicial rulings, and policy documents, but these resources often have significant issues: first, most legal data lack proper structuring and standardization, making it difficult to process; second, many legal documents contain private or sensitive information, limiting their availability to the public; and third, the quantity and diversity of data remain insufficient to support large-scale model training.

This data scarcity directly hinders the development of AI-driven legal systems in China. For deep learning models, especially large language models, the quantity and quality of training data are crucial factors influencing their performance. This problem becomes particularly apparent in legal tasks, such as judicial reasoning and legal interpretation, where the lack of sufficient high-quality data prevents current models from achieving practical performance. Therefore, addressing the shortage of legal data, especially through data synthesis, is the key to enhance the effectiveness of legal AI systems.

As large language models have matured, many legal AI systems have begun to take advantage of their powerful text generation and comprehension abilities. However, their application in the legal domain faces unique challenges. Legal texts require models to generate precise and rigorous language, but large language models often fail to guarantee the accuracy and strictness required in legal documents. For example, models might generate common-sense errors when interpreting legal provisions or misinterpret the intent behind legal rulings. Additionally, the high level of abstraction in legal language may hinder models from

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https://doi.org/10.1016/j.knosys.2026.116267
Received 5 May 2025; Received in revised form 13 April 2026; Accepted 18 May 2026
Available online 27 May 2026
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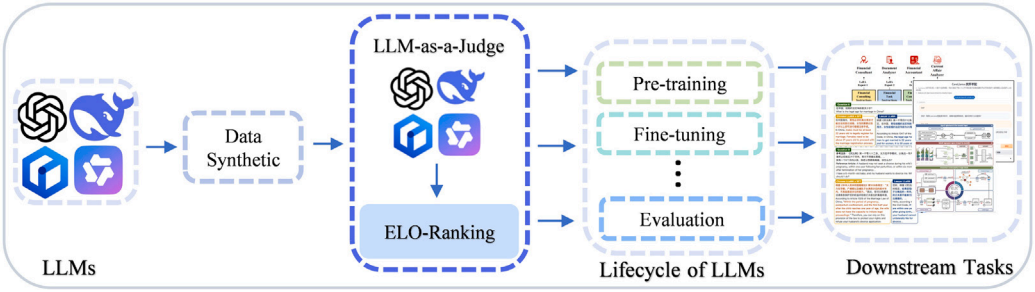


Fig. 1. In the prevalent data synthesis schemes of current large language models, the process generally follows the steps within the gray dashed-line box. Our proposed scheme, however, incorporates an additional component, which is enclosed in the purple dashed-line box. Specifically, we introduce a quality assessment module that combines the LLM-as-a-Judge technique and the ELO-Ranking technique. This addition enables a more effective evaluation of the quality of synthetic data, enhancing the overall data synthesis process for LLMs.

fully capturing the subtle nuances of legal provisions, leading to outputs that lack legal applicability.

These challenges limit the general applicability of large language models in the legal field, and the key to addressing them lies in constructing a more specialized and high-quality legal dataset. In this context, data synthesis techniques have become a promising avenue for research.

Data synthesis has garnered considerable attention as a means to augment training datasets, particularly when data is scarce. Many studies and applications have attempted to generate synthetic data using large language models to expand training sets. However, existing data generation methods face two major issues: first, data generated via simple prompt engineering often lacks quality assurance, leading to noisy outputs, especially in highly specialized domains like law, where the generated text may fail to meet legal requirements; second, existing techniques often lack effective control over data quality, resulting in significant discrepancies in accuracy and consistency. Thus, improving the quality of synthetic data, particularly in the legal domain, remains an open challenge.

Numerous studies have attempted to solve the issue of data synthesis and have achieved preliminary results in various fields. Some international studies have used prompt-based approaches to generate legal data. However, these methods encounter significant challenges when applied to Chinese legal texts due to the complexities of the domain. Moreover, although there are domestic efforts exploring legal data synthesis, most methods still rely on simplistic generation strategies that fail to ensure high-quality outputs. More importantly, there is a lack of targeted solutions for generating Chinese legal texts, particularly those tailored to the unique requirements of the Chinese legal system. Therefore, we propose a method that improves the quality of the synthesized data. As shown in Fig. 1, our approach integrates a quality assessment module consisting of the LLM-as-a-Judge and ELO-Ranking techniques into the traditional data synthesis process of large language models. This module, highlighted in the purple dashed-line box in the figure, is designed to effectively evaluate and enhance the quality of synthetic data. In contrast, the common data synthesis flow in existing LLMs is depicted within the gray dashed-line box. By adding this innovative module, our method addresses the limitations of previous approaches and aims to better meet the specific needs for generating highquality Chinese legal texts.

The main contributions of this work are threefold:

- To the best of our knowledge, this work presents the first automated data synthetic framework for generating high-quality retrieval-augmented generation data tailored specifically for the legal large language models. Our data synthetic method provides high-quality question-answer pairs that are highly relevant to the legal domain, which improves the efficiency of the retrieval module and the accuracy of the generation module. By supplementing training data in this way, the framework addresses the issue of data scarcity in the legal field, boosting the overall performance of legal large language models with the RAG-based technique.
- We introduce an innovative approach in the automated data synthetic process by combining LLM-as-a-Judge with ELO-Ranking technology. In this method, large language models are employed as evaluators through prompt engineering to assess and compare multiple generated paired answers to a given legal question. The model acts as a "judge" determining the relative quality of each pair. Subsequently, we apply the ELO-Ranking technique to rank the generated answers based on their overall relative quality, reducing noise and ensuring the selection of the most accurate and reliable answer, thereby enhancing the overall quality of the synthetic legal data.
- Extensive experiments are conducted to validate the effectiveness of our proposed method through the well-known legal evaluation dataset LawBench. The experimental results demonstrate that our approach can automatically produce high-quality RAG synthetic data without human supervision, making it effective for training trustworthy legal AI models. Additionally, we make the datasets and implementations available to the research community to facilitate further research.[1]

2. Related work

2.1. Data synthesis and augmentation driven by LLMs

In the current research landscape, the utilization of LLMs for data synthesis has emerged as a critical solution to the impending crisis of data scarcity, where the growth rate of high-quality data lags significantly behind the rapid expansion of training datasets. Traditional data synthesis and augmentation techniques have proven inadequate to meet these demands [1–4]. Leveraging LLMs for data synthesis is of substantial significance, as it reduces the reliance on manually curated datasets and generates high-quality, diverse data that meets the functional and lifecycle requirements of LLMs [5–7]. This, in turn, enhances model performance and generalization capabilities, thereby advancing their applications across various domains [8–12] .

Current methodologies for data synthesis using LLMs can be broadly categorized into two approaches: data augmentation and data synthesis [13]. In the realm of data augmentation, techniques include data annotation, where LLMs' linguistic comprehension capabilities are employed to label vast amounts of unannotated data, thereby expediting data preparation in fields such as cross-lingual processing and multimodal learning. For instance, ChatGPT-4 has been utilized to annotate political tweets [14]. Another technique is data reconstruction, which involves transforming and reorganizing existing data

[1] https://github.com/Huijigo/sys-LawLLM.git

to generate additional samples. An example is Disco, which leverages LLMs to produce large-scale, high-quality counterfactual data [15]. Collaborative annotation is also employed, combining the strengths of human annotators and LLMs to allocate annotation tasks based on the uncertainty of LLM-generated labels, as exemplified by the CoAnnotating framework [16].

In the domain of data synthesis, methods include general model distillation, where powerful general-purpose models like ChatGPT and GPT-4 are used to generate datasets that enhance the capabilities of weaker models [17]. A typical example is the Phi-1 series [18]: it leverages GPT-3.5 (a powerful general-purpose model) to generate two core types of synthetic data—Python "textbook data" (< 1B tokens, combining natural language explanations and code snippets) and Python "exercise data" (~180M tokens, consisting of docstring-to-code tasks)—which, together with filtered real code data, form a "textbook-quality" training corpus. With this corpus, the 1.3B-parameter Phi-1 model achieves 50.6% pass@1 on HumanEval and 55.5% on MBPP, outperforming similar or larger-scale models despite minimal training resources. Domain-specific model distillation focuses on generating data for particular fields, such as code programming and mathematics, using domain-tailored models [19–21]. Additionally, model self-improvement techniques enable models to generate high-quality data autonomously to enhance their own performance, as seen in Self-Instruct [22], which automatically generates and optimizes instructional data. These methodologies provide effective pathways for the use of LLMs in data synthesis, playing a key role in the ongoing development of LLMs.

While the preceding approaches have substantially advanced the state of automated data generation, an orthogonal line of research has focused on refining the evaluation of synthetic outputs through the integration of human feedback. Reinforcement Learning from Human Feedback (RLHF), as introduced by Ouyang et al. [23], has demonstrated remarkable efficacy in aligning model behavior with nuanced human preferences by training a reward model on comparative judgments. Recent work has extended this paradigm to specialized domains, including the alignment of crowd-sourced feedback for code generation tasks [24]. Within the context of legal data synthesis, RLHF could, in principle, be deployed to iteratively refine an LLM-based evaluator, equipping it to better discern subtle doctrinal errors, interpretive nuances, or jurisdictional variations that are difficult to capture through static prompting alone. Nevertheless, the practical application of RLHF to large-scale data curation in the legal domain faces substantial obstacles—most notably, the prohibitive expense of expert legal annotation and the inherent variability of legal interpretation across qualified practitioners. These considerations motivate a deliberate design choice in the present work: rather than embedding human feedback directly into the evaluation loop, we concentrate on establishing a fully automated, zero-human pipeline for quality assessment. Our framework accordingly employs a static, prompt-optimized LLM-as-a-Judge coupled with ELO-Ranking to provide a scalable and cost-effective baseline for filtering synthetic legal question-answer pairs. As we demonstrate empirically in Section 4.4, even this static evaluation mechanism yields substantial performance improvements, thereby establishing a robust and practical foundation upon which future human-in the loop refinements—including RLHF may be economically layered.

2.2. Legal knowledge representation and retrieval augmented generation

Recent efforts in legal artificial intelligence have increasingly emphasized the integration of structured legal knowledge—particularly knowledge graphs (KGs)—with large language models to mitigate hallucination and improve reasoning fidelity. A practical approach by Dang et al. (2023) leverages knowledge graphs to represent the semantic essence of legal texts, enabling more precise legal query understanding and response grounding [25]. Building upon this, Dung et al. (2025) proposed a knowledge graph-based legal query system that synergistically combines RAG with LLMs, enhancing both the relevance and traceability of legal information retrieval by anchoring LLM outputs in structured legal entities and relations [26] This graph-based retrieval-augmented paradigm has been proven effective in domain-specific fact-checking tasks to mitigate hallucination and improve reasoning fidelity [27], further validating the value of integrating structured knowledge graphs with LLMs in high-stakes professional domains like law. Complementing these system-oriented architectures, Gray et al. (2024) introduced a novel methodology to automatically discover legal factors, which are the foundational conceptual elements in legal reasoning, directly from case texts using LLMs, without relying on pre-defined ontologies or expert-curated lists [28]. This work demonstrates the potential of LLMs not only as generators but also as tools for unsupervised legal knowledge extraction and conceptual modeling, thereby enriching downstream KG construction.

Furthermore, Siino et al. (2025) conducted a comprehensive literature review of LLM applications in law, identifying key trends across legal document analysis, case outcome prediction, contract review, and legal question answering [29]. Their survey highlights knowledge integration – via knowledge graphs, RAG, or hybrid reasoning – as a critical direction for improving the reliability, interpretability, and domain alignment of legal LLMs, especially in high-stakes professional settings.

In addition, recent studies have provided a broader perspective on the intersection of big data and AI in legal analytics. For example, J Big Data (2024) presented a global bibliometric analysis of AI-driven evidence analysis, identifying key trends, accessibility challenges, and emerging directions in data-driven legal research [30]. This work underscores the growing importance of integrating large-scale data analytics with AI methodologies in legal applications, aligning with our focus on enhancing legal knowledge representation and retrieval through LLM-based data synthesis.

2.3. Recent advances in legal large language models

With the rapid development of large language model technology, the field of legal technology has undergone a profound transformation, leading to the emergence of legal-specific large language models [31–34]. These models are built upon the general framework of LLMs, but are fine-tuned through secondary training on extensive legal datasets (such as legal documents, judicial cases, laws, and regulations.). By incorporating specialized legal knowledge, they are equipped to deal with complex legal tasks effectively.

Since the release of ChatGPT [35] by OpenAI, large language models based on the Transformer [36] architecture have become one of the core directions in artificial intelligence research. These models are widely applied in the field of NLP and have achieved remarkable results in tasks such as machine translation, text generation, and sentiment analysis. With continuous technological advancements, major research institutions and companies have introduced their own base models. Notably, in the English language domain, the LLaMa series (Meta) [37,38] has garnered significant attention for its robust language understanding and generation capabilities, making a considerable impact. Meanwhile, in the Chinese language domain, models like Baichuan (Zhiyuan Institute) [39] and ChatGLM (Tsinghua University) [40,41] have gradually become key tools for tasks in Chinese natural language processing, due to their exceptional performance in the Chinese context.

In the legal domain, the research and application of large language models are steadily advancing, particularly in the construction and optimization of Chinese legal models. In recent years, several domestic teams have successfully launched influential legal large models covering a wide range of legal fields and application scenarios. ChatLaw [42], based on ChatLAW-13B, utilizes diverse data sources such as academic articles, regulations, and judicial interpretations. Through data cleaning and augmentation, it has constructed a high-quality training dataset. HanFei [43], on the other hand, relies on a

vast corpus of cases, regulations, and legal consultation data, combining general Chinese instructions with legal Q&A data to provide accurate answers to legal questions. Lawyer LLaMa focuses on the legal examination domain, utilizing the Chinese National Unified Legal Professional Qualification Examination question bank and the JEC-QA dataset to offer efficient exam preparation support for legal practitioners. LexiLaw [44] integrates legal Q&A data, landmark cases, and legal regulations to build a comprehensive legal knowledge base, enhancing the model's performance in legal reasoning and intelligent Q&A tasks. LawGPT_zh [45] and LawGPT [46], based on ChatGPT [35] and Chinese-Alpaca-Plus-7B [47], integrate a vast array of classic cases, judicial documents, legal exam data, and relevant laws and regulations, providing solid data support for the model's legal applications in contextual dialogue, legal Q&A, and legal clause reasoning.

In terms of data synthesis, these models commonly employ various methods to enhance the quality and diversity of the training data. For instance, ChatLaw [42] optimizes the quality of legal document data through data augmentation techniques, while HanFei [43] enhances its model's adaptability in legal contexts by incorporating a large volume of legal instructions and Q&A data. Lawyer LLaMa generates targeted exam-related situational Q&A by integrating legal exam question bank data. LexiLaw [44] and LawGPT [46] further enrich the models' legal knowledge by using open datasets such as LaWGPT and the Judgment Document Network, as well as proprietary legal datasets.

Although existing large language models have achieved remarkable success in various fields, they still face significant challenges when dealing with complex tasks, particularly in terms of data quality and quantity. Specifically, although models like the LLaMa series [37,38], Baichuan series [39], and ChatGLM series [48] have been pretrained on large volumes of high-quality public data, these models are still limited by the scope and diversity of their training data. This is especially evident in specialized domains such as law and medicine, where there is often a lack of sufficient high-quality domain-specific data. As technology has advanced, synthetic data generation has gradually emerged as one of the key research directions to address these challenges. Through intelligent synthesis techniques, researchers can generate more precise, diverse, and high-quality data tailored to specific domains or tasks, compensating for the shortcomings in existing datasets.

3. Methodology

3.1. Data synthesis

This study proposes an innovative synthetic data generation framework comprising two key components: Response Generation and Quality Evaluation. Specifically, we first employ structured prompt engineering to guide large language models in producing initial datasets. Subsequently, we implement a dual-layer filtration system that integrates an LLM-based automated evaluation mechanism (LLM-as-a-Judge) with an enhanced ELO-Ranking system, enabling iterative quality optimization of the synthesized data.

3.1.1. Response generation

In the Response Generation module, this study primarily utilizes large language models for the generation of legal data through prompt engineering (As shown in Table A.1). Specifically, we first collect a set of legal provisions D from the Chinese legal domain. For each legal provision D_i in D, we input it into the LLM and carefully design a prompt that guides the model to generate questions Q_i highly relevant to the corresponding provision D_i. This process results in a set of (D_i, Q_i) pairs.

Subsequently, each (D_i, Q_i) pair is fed back into the LLM with a newly crafted prompt, instructing the model to generate multiple answers, thus producing a set of potential answers $A_1, A_2, \ldots, A_n$. The final output is a set of $(D_i, Q_i, \{A_1, A_2, \ldots, A_n\})$ triples.

To facilitate evaluation, the generated multiple answers for each question are paired into all possible two-answer combinations, forming the "matches" that will be assessed in the Quality Evaluation module. This approach ensures diversity in candidate answers while enabling systematic pairwise comparison for subsequent ranking.

3.1.2. LLM-as-a-judge

In the legal domain, evaluating data quality and model performance is of paramount importance. To enhance the accuracy and reliability of synthetic legal data, we propose applying the "Large Language Model as a Judge (LLM-as-a-Judge)" paradigm for legal data evaluation and systematically explore its implementation across legal tasks. This approach leverages large language models as evaluative agents to rank and assess generated legal responses, thereby improving data quality.

In our framework, a "match" is defined as a pair of candidate answers generated for the same legal question. For each question, all possible answer pairs are systematically generated, and each pair constitutes a match. The LLM-as-a-Judge evaluates each match using structured prompts (As shown in Table A.2), considering multiple criteria, including question-document relevance, question formulation, citation of legal provisions, and the appropriateness and completeness of the answers. The model first produces a detailed analysis for each match and then outputs a final comparative judgment in a strict format ("id1 > id2", "id1 = id2", or "id1 < id2"), preserving input order. This process ensures reproducibility, transparency, and fairness in evaluation.

Given the specificity of legal tasks, domain-specific adaptations are essential. Legal questions often involve precise terminology, complex procedural knowledge, and nuanced interpretations of statutes and precedents. To improve the model's comprehension of legal data structures and reasoning frameworks, we employ tailored prompts that incorporate legal corpora and case studies. These customized prompts help the model capture legal language patterns and enhance performance in legal reasoning tasks.

For synthetic data evaluation, we adopt a pairwise comparison methodology. When presented with conflicting legal interpretations or arguments – for instance, divergent analyses of tort liability clauses – the model assesses which interpretation demonstrates greater legal validity. Through careful prompt engineering, the model considers statutory consistency, precedent relevance, and logical coherence to produce comparative judgments. In intellectual property disputes, for example, the model evaluates competing patent claims by analyzing the legal basis, evidentiary support, and alignment with established case law, thereby identifying the more legally sound claim.

The complexity and technicality of legal language may lead the model to generate verbose or jargon-heavy responses. To address this challenge, we fine-tune the pre-trained model using simplified legal language exemplars and design prompts to produce more concise and accessible outputs. Our evaluation framework emphasizes not only legal accuracy but also clarity and conciseness, ensuring that legal texts remain precise while enhancing comprehensibility.

During legal judgment tasks, the model may exhibit position bias, favoring arguments presented first or last. To mitigate this bias, we implement a position-swapping protocol and incorporate expert legal review, where professional lawyers audit the model's judgments to identify and correct potential biases. In legal advisory evaluations, we systematically randomize answer sequences and apply appropriate adjustments to ensure comprehensive and impartial assessment.

To enhance transparency, we document the model's reasoning process and decision-making rationale in detail. These records provide valuable references for legal practitioners to assess the reliability of model outputs. Furthermore, transparency in the reasoning process allows legal experts to understand the model's decision logic, ensuring that evaluation outcomes adhere to legal standards and can withstand rigorous scrutiny.

All specific prompt designs and implementation details are documented in the Appendix to ensure reproducibility and facilitate further research by both academic and legal professionals.

3.1.3. ELO-ranking

After utilizing the LLM-as-a-Judge methodology to generate multiple comparative results, the next step in the Quality Evaluation module is to apply the ELO algorithm. This algorithm ranks the responses generated by the large language model for the same legal question based on pairwise comparisons. The ELO-Rining system, originally designed to rank chess players based on their game outcomes, is well suited for evaluating competitive entities — in this case, legal responses generated by the large language model — and determining relative quality through pairwise comparisons.

(1) ELO Algorithm Overview The ELO-Ranking system works by comparing pairs of entities and assigning them scores based on their relative performance. After each comparison, the rankings are updated to reflect the performance of the entities in relation to each other. The fundamental principle is that the stronger entity (the one that wins in the pairwise comparison) gains points, while the weaker entity loses points. The magnitude of points exchanged depends on the difference in score between the two entities being compared.

In our adaptation for the legal domain, the "entities" are the legal responses generated by the LLM-as-a-Judge. These responses are compared pairwise to determine which response is more legally valid or accurate, based on the evaluation criteria established in the previous section, such as statutory consistency, relevance to precedent, and logical coherence.

(2) Application of ELO-Ranking to Legal Data

Pairwise Comparisons: After generating n candidate answers for a specific legal question, we create A_n^2 distinct pairwise comparisons. For each comparison, the model evaluates which response better addresses the legal issue, considering the legal criteria mentioned above.

ELO Score Calculation: Following the pairwise comparisons, the ELO-Ranking system is applied to rank the responses. Each response starts with an initial score (e.g., 1500, similar to the starting point in chess). For each pairwise comparison, the expected outcome is calculated as:

$$\mathcal{E} = \frac{1}{1 + 10^{\frac{R_{\text{opponent}} - R_{\text{player}}}{400}}}$$

where R_{player} and R_{opponent} represent the current scores of the two responses. Based on the actual result of the comparison (whether a response wins or loses), the ELO scores are updated using the formula:

$$\mathcal{R}_t' = \mathcal{R}_t + \mathcal{K}_t \times (\mathcal{S}_t - \mathcal{E}_t)$$

where R' is the updated score, S is the actual result (1 for a win, 0 for a loss), and E is the expected result.

Ranking and Final Selection: After the A_n^2 pairwise comparisons, the responses will have their ELO scores adjusted accordingly. The response with the highest ELO score is considered the most accurate or legally valid answer. This ranking process ensures that the selected response is not only the most legally sound but also superior in multiple aspects, such as clarity, conciseness, and logical coherence.

(3) Benefits of Using ELO for Legal Data Ranking

Dynamic Adjustments: The ELO-Ranking system allows continuous updates to the rankings based on ongoing evaluations, making it an adaptive and robust method for refining the quality of synthetic legal data as more comparisons are conducted.

Contextualized Decision-Making: ELO not only takes into account the "win" or "loss" from each comparison but also adjusts for the relative strength of the responses. This contextual approach helps prevent extreme ranking shifts caused by randomness or biases inherent in a single comparison.

Transparency: The iterative nature of the ELO scoring system allows for a transparent process for ranking responses. Researchers and legal professionals can track how the score of each response evolves over time, providing insights into the factors influencing its ranking.

(4) Addressing Potential Biases

Although ELO provides an effective ranking mechanism, it is crucial to address potential biases in the process. Specifically, the sequence in which pairs are compared may impact the final ranking. To mitigate this:

- **Randomization:** We randomize the order of pairwise comparisons to reduce positional bias.
- **Bias Monitoring:** We monitor the ELO scores and adjust for any patterns that may indicate bias related to specific legal contexts or argument types. If necessary, we apply corrective measures to ensure fair and balanced rankings.

(5) Adaptation of the ELO-Ranking System to Legal QA Evaluation

To adapt the game-theoretic ELO system to the task of ranking legal question–answer (QA) pairs, several domain-specific modifications were introduced to ensure accurate, fair, and interpretable evaluation within the legal context.

Task Mapping: In our framework, each candidate legal answer generated for the same question is treated as a "player" in the ELO system. The "matches" correspond to pairwise comparisons performed by the LLM-as-a-Judge, which determines which answer demonstrates higher legal validity and reasoning quality. Unlike traditional competitive games, these outcomes are derived from structured evaluations of legal soundness rather than random events, reflecting factors such as statutory consistency, precedent relevance, and logical coherence.

Soft Scoring Scheme: To better capture fine-grained quality differences, we extend the conventional binary win/loss outcome in ELO to a continuous scoring mechanism, where the comparison result $S_t \in [0, 1]$ represents the confidence margin of superiority between two answers as judged by the LLM. This adjustment allows the ELO updates to reflect nuanced distinctions in legal reasoning and factual accuracy.

Bias and Fairness Control: To mitigate potential order or contextual biases during pairwise comparisons, we randomize the sequence of matchups and monitor the score distributions across different legal subdomains. When systematic imbalance is detected, normalization adjustments are applied to maintain fairness and stability in the ranking process.

Domain-Specific Dynamic Adjustment: During score updates, the K-factor is dynamically scaled based on the complexity and semantic richness of the legal question. Complex multi-clause or interpretive cases receive smaller update magnitudes to preserve ranking stability, while simpler factual queries are updated with larger step sizes to accelerate convergence.

These adaptations ensure that the ELO-based ranking framework not only preserves its theoretical rigor but also aligns with the requirements of legal QA evaluation, enabling robust, transparent, and domain-sensitive quality assessment of synthesized data.

3.1.4. Data collection

When applying large language models to vertical domains, a common approach is to combine fine-tuning with Retrieval-Augmented Generation. Based on this approach, this paper adopts the method of fine-tuning large language models to enhance the model's expertise in the legal domain while optimizing its language style and expression.

In the RAG framework, determining when to perform retrieval is a critical issue, as it directly affects the model's reasoning capabilities and the credibility of the generated results. To address this challenge, this study systematically modifies the fine-tuning dataset, which was constructed through automated synthesis and evaluation, using the following methodology:

First, an expert review mechanism is employed to assess fine-tuning data in the legal domain, evaluating whether the response to each data point requires the use of professional legal statutes. For samples that indeed necessitate legal knowledge, we first extract the relevant legal statutes. We then insert a ⟨Retrieval⟩ tag before the legal statutes in the response to simulate the retrieval process: two highly relevant legal

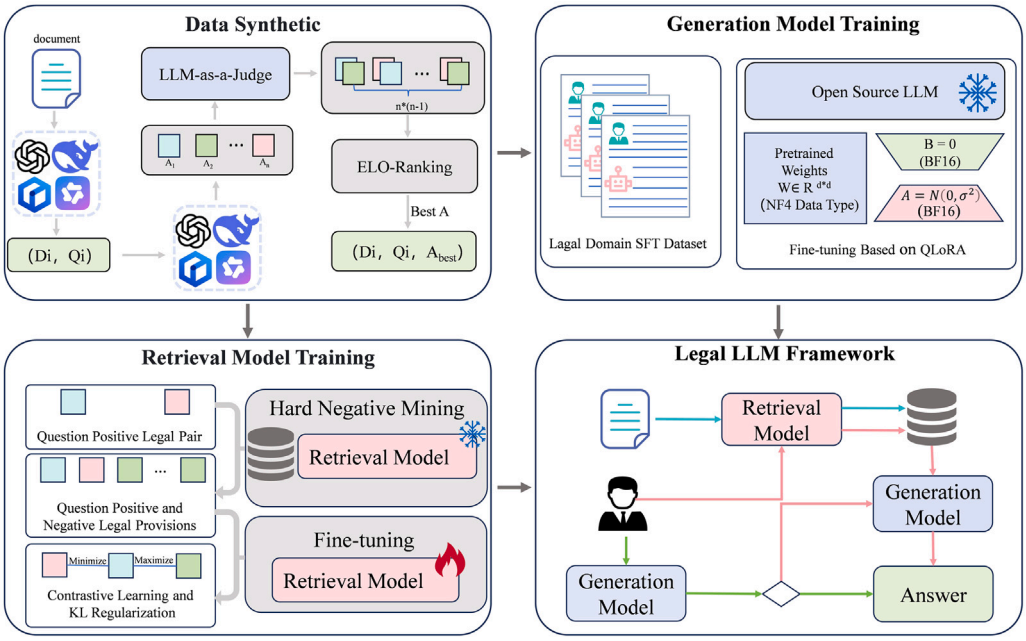


Fig. 2. Framework Diagram. The QA pairs generated from the Data Synthetic method are used for training the retrieval model, and the best of generated dialogue data is used for training the generation model. In the Legal large language model module, → represents the process of storing legal provisions into the vector database through the retriever, → represents the steps where the model does not require retrieval, and → represents the process of Retrieval-Augmented Generation.

statutes are introduced alongside two randomly selected statutes from the legal knowledge base as distractors, with the process ending with a ⟨/Retrieval⟩ tag. This design aims to simulate the complete process of invoking a retriever in practical applications.

After the model generates a response, we explicitly annotate the specific legal statutes referenced using ⟨Reference></Reference⟩ tags, thus ensuring traceable and reliable generated results. This annotation mechanism not only enhances the credibility of the model's output but also facilitates subsequent model interpretation and error analysis.

Through this systematic data modification approach, we have constructed an effective fine-tuning dataset that enables the model to determine the appropriate retrieval time, laying a crucial foundation for improving the application of the RAG framework in the legal domain.

Furthermore, to better support the application of retrieval-augmented techniques in the legal vertical domain, this study has additionally curated a high-quality dialogue-form dataset from a large volume of legal documents, legal competition datasets, and judicial exam questions. These data points are not synthetic and have strong credibility. They are also modified following the same logic used for the synthetic data, forming an integral part of the fine-tuning dataset.

3.2. Framework

Through the method of data synthesis, this study constructs an automated data synthetic process to train the retrieval model and the generation model, and builds a large language model enhanced by active retrieval in the legal field. The framework diagram is shown in Fig. 2.

3.2.1. Generation model fine-tuning

After completing data collection and preprocessing, we employ a parameter-efficient fine-tuning method based on QLoRA [49] to conduct Supervised Fine-Tuning (SFT) on the generate LLM. The supervised fine-tuning scheme in this study aims to achieve three core objectives: (1) to adapt the model to response generation for downstream tasks in the legal domain; (2) to guide the model in autonomously deciding the timing of retrieval by generating the ⟨Retrieval⟩ tag to trigger knowledge retrieval; and (3) to enhance the model's understanding and application of legal provisions, accurately associating generated content with relevant legal bases.

For samples that do not require legal retrieval, the input sequence is defined as $X = [x_1, x_2, \ldots, x_n, x_{n+1}, \ldots, x_{n+m}]$, where:

- The first n tokens $[x_1, x_2, \ldots, x_n]$ constitute the system prompt and user question segments.
- The subsequent m tokens $[x_{n+1}, \ldots, x_{n+m}]$ constitute the standard response segment.

The model achieves loss focus through a positional masking mechanism: a mask vectorM = $[m_1, \ldots, m_{n+m}]$ is defined, where:

$$m_t = \begin{cases} 0 & t \le n \text{ (question segment)} \\ 1 & t > n \text{ (answer segment)} \end{cases}$$

The cross-entropy loss function is adjusted as:

$$\mathcal{L}_t = -\sum_{i \in \mathcal{V}} m_t \cdot y_{t,i} \log(\hat{y}_{t,i})$$

where V denotes the vocabulary of the tokenizer, $y_{i,t}$ represents the one-hot vector encoding of the ground truth label, and $\hat{y}_{t,i}$ corresponds to the predicted probability distribution of the model at the current time step. This design ensures that the model optimization is solely focused on the quality of answer generation, avoiding interference from question segment tokens during training.

For samples that require legal retrieval, the input sequence is extended to:

$$X = \Big[\underbrace{x_1, \ldots, x_n}_{\text{question segment}} \; \underbrace{x_{n+1}, \ldots, x_{n+\tau}}_{\text{response segment}}, \underbrace{r_1, \ldots, r_{t_1}}_{\langle\text{Retrieval}\rangle}, \underbrace{\ell_1, \ldots, \ell_k}_{\text{laws segment}}, \underbrace{x_{t_2}, \ldots, x_{n+m}}_{\text{response segment}} \Big]$$

A quadruple masking strategy is implemented:

- Question segment mask: This segment should be masked with zeros, just like the samples that are not required to be retrieved.

- Response segment mask: The response segment of the large language model should be set to 1, indicating that the model can correctly respond to the user's query and accurately utilize the retrieved legal provisions to answer the user's question.
- Legal provision segment mask: The provisions $\ell_1, \ldots, \ell_k$ obtained by the simulated retriever are set to zero to prevent the model from memorizing legal knowledge.
- Dynamic retrieval tag supervision: The model is forced to accurately predict the position of the ⟨Retrieval⟩ tag in the answer segment.

The loss calculation is improved as:

$$\mathcal{L}_t = -\sum_{i\in\mathcal{V}} \mathbb{I}_{\text{valid}}(t)\cdot y_{t,i}\log(\hat{y}_{t,i}),$$

$$\mathbb{I}_{\text{valid}}(t) = \begin{cases} 0 & t \in [0,n]\cup(n+\tau+t_1, n+\tau+t_1+k] \\ 1 & t \in [n+1, n+\tau]\cup(n+\tau+t_1+k, n+m] \end{cases}$$

This scheme achieves joint optimization of retrieval decision-making and knowledge application, ensuring both the model's autonomous judgment of retrieval needs and the accuracy of legal provision citations.

3.2.2. Retrieval model fine-tuning

This section of the study is dedicated to optimizing the domain-specific performance of the retrieval model, with the objective of improving its ability to retrieve legal knowledge while preserving its general semantic understanding capabilities. To achieve this, we employ the FlagEmbedding framework, which integrates a hybrid training strategy that combines contrastive learning and knowledge distillation. The training process comprises two primary stages: the construction of training data and the optimization of the model.

In the training data construction phase, for a given query-positive sample pair (q, d^+), an enhanced training set is generated by performing k-nearest neighbor retrieval using the teacher model on a domain-specific legal corpus C:

$$D_q = \{(q, d^+, d_1^-, \ldots, d_k^-, s_{qd^+}, s_{qd_1^-}, \ldots, s_{qd_k^-})\}$$

$$N_d = \{d_1^-, \ldots, d_k^-\}$$

where the negative sample set N_d are generated as follows:

$$N_d = \underset{d\in C\setminus\{d^+\}}{\text{argtopk}}\, \mathcal{M}_\theta(d^+, d)$$

Model optimization employs a hybrid loss function design to balance representation transfer and knowledge retention. The contrastive loss term is defined as:

$$\mathcal{L}_{\text{contrastive}} = -\log \frac{e^{s(q,d^+)/\tau}}{e^{s(q,d^+)/\tau} + \sum_{i=1}^{k} e^{s(q,d_i^-)/\tau}}$$

where $s(\cdot,\cdot)$ denotes the cosine similarity function, and τ is the temperature hyperparameter. To mitigate catastrophic forgetting, a KL regularization term is introduced:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}\left(p_{\mathcal{M}}(q,d) \parallel p_\theta(q,d)\right)$$

where p_θ and $p_{\mathcal{M}}$ represent the probability distributions of similarity of the fine-tuned model and the teacher model, respectively. The final objective function is as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{contrastive}} + \lambda\mathcal{L}_{\text{KL}}$$

This joint optimization strategy effectively enhances domain-specific retrieval performance while maintaining the general semantic understanding capabilities of the base model.

4. Experiments

4.1. Evaluation metrics

LawBench is a comprehensive evaluation index system designed for the systematic assessment of generative large language models within the framework of China's legal system. This evaluation framework is grounded in legal cognitive theory and is structured around three core dimensions: Memory, Comprehension, and Application of legal knowledge. Specifically, the framework encompasses 20 evaluation tasks, each equipped with 500 rigorously selected evaluation data points, culminating in a high-quality dataset of 10,000 entries. Through this multi-dimensional evaluation system, it is possible to comprehensively and systematically assess the cognitive capabilities and professional proficiency of generative large language models in the legal domain, thereby providing a scientific and reliable benchmark for the advancement of legal artificial intelligence.

4.2. Experimental settings

We conducted model training on a server equipped with dual 80G GPUs using the PyTorch framework. For generative tasks, we employed Meta's open-source Meta-Llama-3-8B-Instruct[2] as the base model, setting the maximum input sequence length to 8192 tokens. During the distributed training process, the batch size per GPU was set to 2, and the model was trained for 2 epochs. We utilized a learning rate of 2×10^{-4} for fine-tuning of QLoRA parameters. For the retrieval model, we selected the BAAI-released pre-trained model bge-zh-large-1.5 [50], which has a maximum input sequence length of 512 tokens. The initial learning rate was set to 1×10^{-5}. For each single piece of data in the training of the retrieval model, for one positive example, we will mine 10 negative documents. During dual-GPU parallel training, the batch size per GPU is set to 32, and the model was trained for 2 complete epochs to optimize the parameters. The hyperparameters of τ and λ are set to 1. For detailed model training details, please refer to the open-source code. During the evaluation process of LawBench, we perform the retrieval and recall of five legal provisions for each ⟨Retrieval⟩ tag response.

4.3. Datasets

In this study, the datasets constructed for the experiments consist of three main types, each serving different purposes and exhibiting distinct characteristics.

- **Provenance-based Legal Dataset**: This dataset is derived from a large-scale corpus that includes millions of judicial rulings, publicly available legal competition datasets, and judicial examination questions. Through a rigorous data cleaning and transformation process, we obtained a dialogue-format dataset consisting of 14,215 entries. These data represent various tasks within the legal domain and are considered highly feasible and authentic, with a solid basis for their construction.
- **Response Generation-Only Synthetic Legal Dataset:** This dataset consists of dialogue data generated exclusively through prompt engineering with a large language model. The data, which includes 6000 entries, were synthesized directly without undergoing any post-generation evaluation or quality assessment. As such, these data represent a purely synthetic collection generated by the model based on the specified prompts, corresponding to the "Response Generation" phase of the framework.

[2] https://github.com/meta-llama/llama3/blob/main/MODEL_CARD.md

Table 1
Effectiveness of data selection pipeline in legal domain.

	Legal knowledge QA	Argument mining	Event extraction	Case analysis
Unevaluated	0.7801	0.5603	0.4782	0.6941
Evaluated	**0.8192**	**0.6253**	**0.7712**	**0.7648**

- **Response Generation and Quality-Evaluated Synthetic Legal Dataset:** This dataset also consists of dialogue data generated using prompt engineering with a large language model, but it undergoes an additional quality evaluation process. Specifically, the data are subject to the dual-layer filtration system, which integrates an LLM-based automated evaluation mechanism (LLM-as-a-Judge) with an enhanced ELO-Ranking system. This process ensures that the data meet specific quality criteria, and the final dataset comprises 6000 entries. These evaluated synthetic data represent a more refined and validated set, aligning with both the "Response Generation" and "Quality Evaluation" components of the framework.

4.4. Preliminary experiment

To evaluate both the performance and domain validity of the synthetic QA data, we conducted a preliminary experiment using LawBench – a comprehensive benchmark for legal reasoning – across four representative tasks: Legal Knowledge QA, Argument Mining, Event Extraction, and Case Analysis.

All synthetic QA pairs were generated from authentic legal texts, including statutes, judicial interpretations, and real-world case documents, ensuring that each question–answer pair is legally grounded and contextually relevant. During evaluation, an LLM-as-a-Judge framework was employed to perform pairwise comparison of generated answers based on four explicit legal criteria: (1) question–document relevance, (2) formulation clarity, (3) citation adequacy, and (4) reasoning completeness. The standardized evaluation prompt is provided in Appendix. The resulting ELO-based ranking mechanism was then used to filter low-quality or legally inconsistent data.

The comparative results, presented in Table 1, demonstrate consistent quality improvements across all legal tasks after filtering. Absolute accuracy gains range from +3.91% (Legal Knowledge QA) to +29.30% (Event Extraction), with relative improvements spanning from 5.02% to 61.33%. Notably, Event Extraction exhibits the largest increase (from 0.4782 to 0.7712), suggesting that our pipeline effectively removes factually or legally noisy samples that hinder structured information extraction. Case Analysis also benefits from improved logical coherence (0.6941 to 0.7648), while Argument Mining shows moderate but stable enhancement. Overall, these results indicate that our filtering pipeline not only improves model accuracy but also enhances the legal soundness and factual reliability of synthetic QA data.

4.5. Baselines

To validate the effectiveness of our proposed method in terms of evaluation metrics, we referred to the relevant results from Lawbench. These results were then compared with those of the not fine-tuned meta/llama3-8B-instruction model, which will be denoted as Llama3 in the following sections. The FT-Llama3 model, as proposed in this paper, is the Llama3 model that has been fine-tuned using the Provenance-based Legal Dataset and the Response Generation and Quality-Evaluated Synthetic Legal Dataset. Here, RAG implies using the fine-tuned retrieval model to assist in the generation process. The comparison results are presented in Table 2.

Table 2
Comparison of the proposed method with the baseline model on the LawBench evaluation metrics.

	Memory	Comprehensive	Application	Total Ave
Llama3	0.2086	0.2548	0.3574	0.2912
chatlaw-13b	0.2163	0.2821	0.4123	0.3276
fuzi-mingcha-7b	0.1651	0.3010	0.4086	0.3305
yulan-chat-2-13b-fp16	0.1872	0.3219	0.3947	0.3376
internlm-chat-7b	0.2720	0.3196	0.3979	0.3462
internlm-chat-7b-8k	0.2793	0.3419	0.3960	0.3573
qwen-7b-chat	0.2627	0.3266	0.4510	0.3700
freewilly2-70b	0.2459	0.4258	0.3869	0.3923
GPT-3.5-turbo-0613	0.2593	0.4331	0.4474	0.4215
FT-Llama3+RAG(Ours)	**0.5346**	**0.4901**	**0.4618**	**0.4832**

4.6. Synthesis data scale analysis

During the data synthesis phase, we adopted a synthesis scheme based on large language models, constructing question-answer pairs by generating questions from documents. Given the potential value of these question-answer pairs in the training of retrieval models, we further explored their application in this context. To this end, we conducted a learning curve analysis using on the retrieval model. This analysis aimed to quantify the impact of synthetic data on the performance of the retrieval model, clarify the effectiveness of synthetic data at different scales, and thereby provide data support for the optimization of retrieval models.

This study constructs an experimental dataset comprising 6000 synthetic samples, partitioned into training and test sets at a 4:1 ratio (4000 for training and 2000 for testing). To investigate the impact of training data scale on model performance, we implement a progressive training strategy. The training set is divided into four incremental subsets at 1000-sample intervals (ranging from 1000 to 4000 samples). Each subset undergoes two training epochs under identical hyperparameter configurations.

The experimental results are shown in Fig. 3. When the training data size gradually increased from 1000 to 2000 instances, all Top-K evaluation metrics showed a significant upward trend. Specifically, the Top-1 recall rate increased from 0.756 to 0.805 (a relative increase of 6.5%), and the Top-20 recall rate increased from 0.9715 to 0.9835 (a relative increase of 1.2%). Notably, all metrics almost reached their peak values when the data size was 2K, and the Top-10 recall rate reached 0.975 (an increase of 2.4 percentage points compared to the initial value), indicating that the model's representation learning ability reached its optimal state at this scale.

When the data size exceeded 2K, the model performance exhibited periodic fluctuations and then stabilized, suggesting that 2K was the performance saturation point. Further increasing the size of the data (3K, 4K) had a limited effect on improving the metrics, which verified the rationality of using 2K as the basic training set. Based on these findings, in this study, we selected 2K synthetic data as the basic training set for the retrieval model and supplemented it with 2K real legal domain data, finally forming a mixed training set of 4K samples to balance the model's generalization ability. In this study, we utilize the RAG annotation to indicate that the generative model employs a retrieval model trained on this 4K dataset for retrieval-augmented generation.

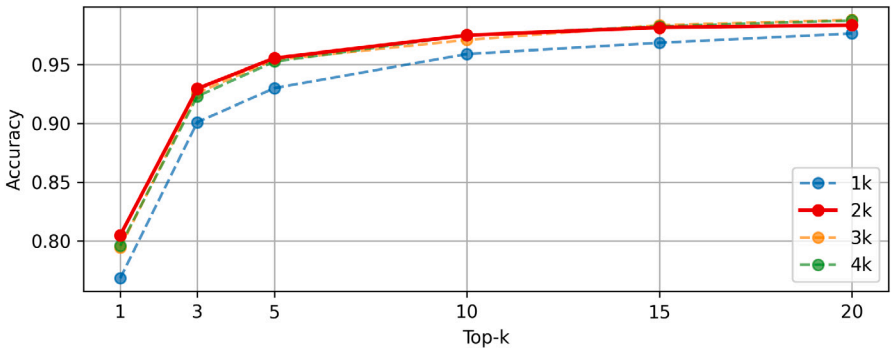


Fig. 3. Learning curve analysis of the retrieval model using synthetic data at different scales.

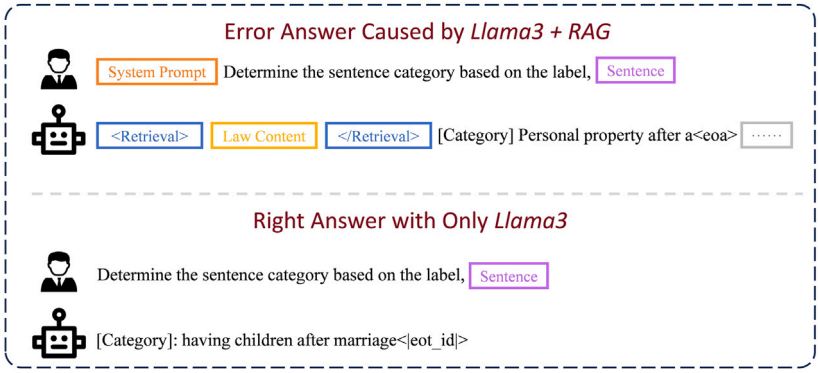


Fig. 4. Comparison of the proposed method with the baseline model on the LawBench evaluation metrics. System Prompt represents the system prompt words for active retrieval. Sentence is the input content of the user's question. <Retrieval > and </Retrieval > enclose the legal provisions retrieved by the retrieval model. Law Content represents the relevant legal provisions retrieved based on the user's question. ..... represents some of the output of the model's question analysis.

Table 3
Overall performance comparison.

	Memory	Comprehensive	Application	Total Ave
Llama3	0.2086	0.2548	0.3574	0.2912
Llama3+RAG	0.4645	0.2316	0.4190	0.3298
FT-Llama3+RAG	**0.5346**	**0.4901**	**0.4618**	**0.4832**

4.7. Overall performance analysis

This study employs a hybrid dataset strategy for model training and evaluation. Specifically, we constructed a fine-tuning dataset comprising 14,215 real data points and 6000 synthetic data points. Additionally, we built a retrieval training set using 2000 synthetic data points and 2000 real-world question document pairs. For the evaluation phase, we performed systematic experiments on 20 tasks based on the Lawbench benchmark. The experimental design includes two main comparison groups: (1) the performance of the Llama3-8B-Instruction model without fine-tuning in both generation and retrieval frameworks; and (2) the performance of the model fine-tuned with our optimal strategy within the RAG framework.

From the experimental results presented in Table 3, we can draw the following conclusions: (1) The RAG mechanism effectively enhances legal memory capabilities but requires fine-tuning to compensate for comprehension deficiencies. For the untuned Llama3 model, the RAG framework improves the legal memory metric from 0.2086 to 0.4645 (+123%). However, the un-tuned model exhibits an 8.3% performance degradation in legal comprehension tasks. Analysis of the results reveals that Llama3 triggered retrieval 4151 times in 5000 legal comprehension tasks, indicating that over-reliance on RAG may lead to misalignment in knowledge dependency. An example of an error in the response caused by retrieval is shown in Fig. 4. (2) Fine-tuning of the joint data significantly enhances the efficiency of legal knowledge application. The comprehensive score of FT-Llam3+RAG (0.4901) improves by 92% compared to Llama (0.2548). In legal application tasks (Application), FT-Llama3+RAG (0.4618) outperforms Llama3+RAG (0.4190) by 10.2%, demonstrating that fine-tuning enables the model to apply legal knowledge more flexibly in real-world scenarios. (3) The "retrieval-fine-tuning" collaborative framework achieves cross-dimensional performance breakthroughs. FT-Llam3+RAG significantly outperforms other models in memory (0.5346), legal comprehension (0.4901), legal application (0.4618), and overall average score (0.4832). The total score increases by 46.5% compared to Llama3+RAG and by 65.9% compared to Llama3, validating the synergistic effect of retrieval enhancement with targeted fine-tuning.

4.8. Ablation study

This study employs an ablation experimental design, systematically evaluating the enhancement effect of synthetic data on model performance through the control variable method within the Lawbench evaluation framework. Specifically, under the condition of a fixed retriever architecture, we quantify the contribution of synthetic data to model performance metrics by conditionally removing the training dataset of the generative model.

Through a systematic analysis of the four experimental datasets, the critical pathways to optimize the performance of legal large language models can be effectively identified, as demonstrated in Fig. 5.

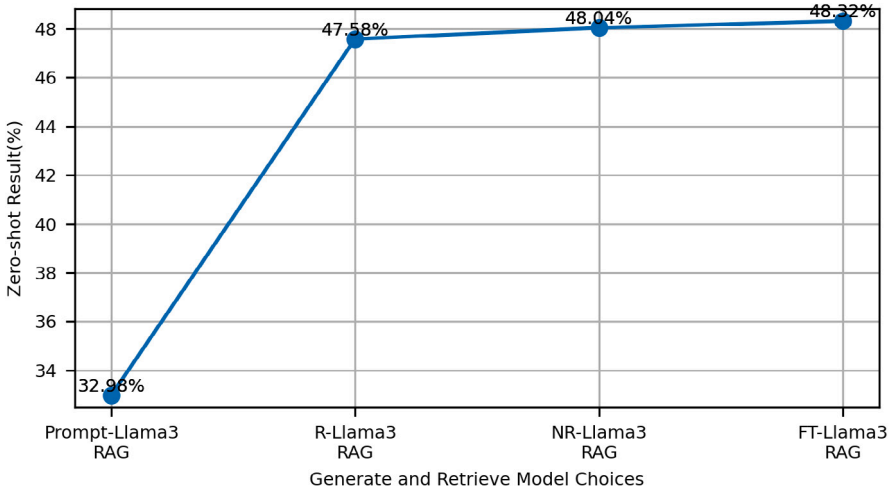


Fig. 5. Comparison of the proposed method with the baseline model on the LawBench evaluation metrics.

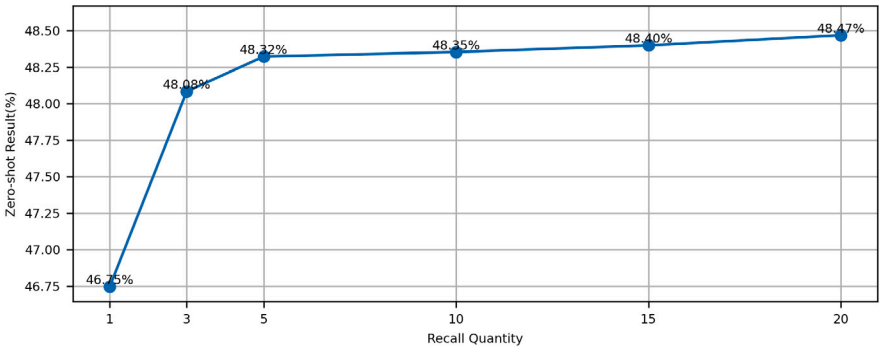


Fig. 6. Top-K recall analysis of our proposed method.

Without fine-tuning, the baseline model Prompt-Llama3, which relies solely on prompt engineering and retrieval augmentation, achieved a score of 0.3298, highlighting the significant limitations of general large models in professional legal scenarios. When fine-tuned with real-world data, the score of R-Llama3 increased to 0.4758, representing a 44.3% improvement. This result validates the necessity of domain-specific training and indicates that the model has preliminarily grasped the characteristics of legal texts and fundamental reasoning patterns. More notably, incorporating synthetic data into the training set led to further performance gains. Using unevaluated synthetic data with real-world data, the NR-Llama3 achieved a score of 0.4804, while quality-assessed synthetic data pushed the score of FT-Llama3 to 0.4832. These two stages contributed improvements of 0.96% and 1.57%, respectively, compared to fine-tuning with only real data. This demonstrates the potential of synthetic data, not only in overcoming the limitations of real data scale but, more importantly, in enhancing model performance through the selection of high-quality samples via carefully designed evaluation mechanisms.

4.9. Top-K recall analysis

In order to further assess the performance of the proposed method, a Top-K recall analysis was carried out to evaluate the model's capacity for retrieving relevant legal provisions. As presented in Fig. 6, the recall rate of this method generally exhibits an upward trend as the recall quantity increases. This implies that the method can progressively enhance its recall performance with the increase in the number of recalls.

The increasing trend of the recall rate with respect to the quantity of the recall in the figure demonstrates that the model possesses excellent information extraction and noise-resistance capabilities. When confronted with a large volume of legal provisions, the model is capable of precisely selecting the applicable content. Consequently, it can effectively improve the accuracy and relevance of the answers.

5. Conclusions

In this paper, we present a novel framework designed to address a critical challenge in legal large language models: the generation of high-quality synthetic data, particularly for domain adaptation. Through systematic experiments and analysis, we draw two key conclusions: First, we propose a dual-stage legal data synthesis framework that combines large language model-based generation with Elo-score-based optimization, establishing a new paradigm for data synthesis in the legal domain. Experimental results demonstrate that synthetic data evaluated using a "LLM-as-a-Judge" and filtered via ElO-Ranking outperform unfiltered synthetic data in terms of performance. Second, by optimizing the collaboration between the generative model and the retrieval model, we achieve state-of-the-art performance on the LawBench benchmark, particularly showing significant advantages in legal understanding and application. This framework not only effectively enhances memory, understanding, and application scores, but also validates the efficacy of our joint training strategy in developing comprehensive legal reasoning capabilities.

CRediT authorship contribution statement

Wenqi Ren: Methodology. **Lixing Shen:** Investigation. **Jiang Zhu:** Supervision. **Shaohui Li:** Validation. **Yinxia Hong:** Visualization. **Jiawei Wang:** Project administration. **Da Cao:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors are highly grateful to the reviewers for their careful reading and insightful comments. The work is supported by the Natural Science Foundation of Hunan Province, China (No. 2022JJ30159 and No. 2023JJ20013), the National Natural Science Foundation of China (No. 61802121), and the Fundamental Research Funds for the Central Universities, China.

Appendix. Supplementary materials

See Tables A.1 and A.2.

Data availability

I have shared the link to my data/code at the manuscript.

Table A.1
Retrieval-based legal question answering system prompt.

Prompt content:
Role: Legal Consulting Expert Background: When users raise legal-related questions in the conversation, accurate legal knowledge is required to provide answers. You are an experienced legal consulting expert with profound legal knowledge and rich practical experience. Tool: You have an external retriever. The ⟨Retrieval⟩ tag you output will call the external retriever, and the external retriever will return the retrieved information enclosed in ⟨Retrieval⟩⟨/Retrieval⟩. Goals: Utilize the external retriever to provide accurate and timely legal consulting services, ensuring that users' questions are effectively answered. Workflow: 1. Understand the legal question raised by the user. 2. Determine whether the question requires external information retrieval. 3. If retrieval is needed, insert the ⟨Retrieval⟩ tag and call the external retriever. 4. Based on the retrieval results, provide a conclusion. After the conclusion, use ⟨Reference⟩⟨/Reference⟩ to enclose the meta_data of the retrieved information used. 5. Continue to determine whether the answer requires further retrieval. If yes, return to Step 2; otherwise, end the answer.

Table A.2
ELO ranking prompt.

Prompt content:
Please act as an impartial evaluator to assess the quality of the following two question-answer pairs generated based on the same document. Your assessment shall be conducted according to the following four criteria: 1. Relevance between the question and the document. 2. Expression form of the question, preferring questions without explicit legal article citations. 3. Appropriateness of legal citation in the answer. 4. Appropriateness and detail of the answer. Assessment requirements: 1. Provide analysis. 2. Strictly compare the two pairs and output one of: - Answer: {{id1}} > {{id2}} - Answer: {{id1}} = {{id2}} - Answer: {{id1}} < {{id2}} Input Format: Legal Basis: {{document}} Pair 1: {{id1}}, {{question1}}, {{answer1}} Pair 2: {{id2}}, {{question2}}, {{answer2}}

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